# The allegory axioms, and what defines them in the Frobenius calculus

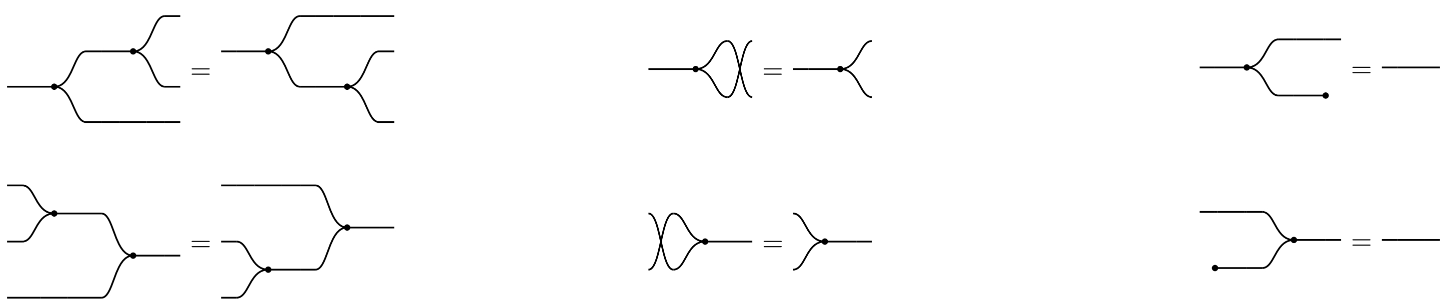
by

## §1. Cartesian bicategory of relations

**Definition 1.1.**
A **cartesian bicategory of relations** is a poset-enriched category(bicategory) that is has a symmetric monoidal product ⊗ (cartisin product of set in Rel, vertical junctposition in the diagram) and

$$\forall \text{ object } A,\ (A,\ ◁,\ ⊸) \dashv (A,\ ▷,\ ⟜),\ \textbf{Frobenius}, \text{ arrows } \textbf{lax}.$$

(◁ : A → A ⊗ A, ⊸ : A → 𝕀) formed a cocommutative comonoid
(▷ : A ⊗ A → A, ⟜ : 𝕀 → A) formed a commutative monoid



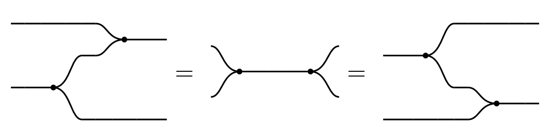
In `Rel(Set)`, the four generators are these relations on a set `A`:

| | |
|---|---|
| ◁ | = {(a, (a, a))} — copy: one in, two identical out |
| ▷ | = {((a, a), a)} — merge: a pair passes only when its two components agree, and the common value comes out |
| ⊸ | = A × {∗} — discard: forget the value, keep only that there was one |
| ⟜ | = {∗} × A — create: any element at all |

### §1.1. ∀ object A, (A, ◁, ⊸) ⊣ (A, ▷, ⟜)

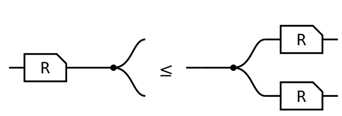
▷ ◁ ≤ 𝟙      𝟙 ≤ ◁ ▷      ⟜ ⊸ ≤ 𝟙      𝟙 ≤ ⊸ ⟜

### §1.2. The Frobenius law

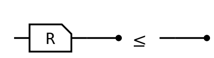


The only clause coupling the comonoid to the monoid beyond adjointness. Both sides reduce to the same picture, ▷ ◁ — merge, then copy — which is the two-in two-out **spider**. Every connected diagram built from these four generators collapses to the spider on its own inputs and outputs, and this equation is what starts that collapse. A **bubble** is the other order, ◁ ▷: copy then merge, a closed loop, which the idempotency of ∩ uses below.

### §1.3. Every arrow is a lax comonoid homomorphism



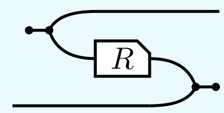

R◁ ≤ ◁(R⊗R)




R⊸ ≤ ⊸

# §2. $^\circ : \mathcal{C}^{op} \longrightarrow \mathcal{C}$ is a 2 functor

**Definition 2.1.**
The **converse** `R°` of `R : a → b` is `R` with both of its wires turned round,

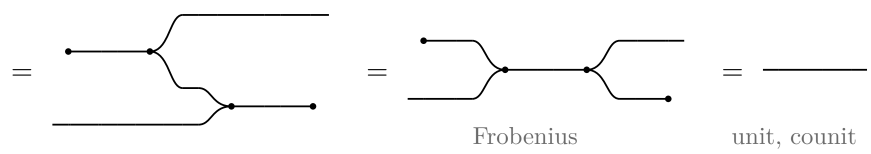



`R° = (⟜◁ ⊗ 𝟙) (𝟙 ⊗ R ⊗ 𝟙) (𝟙 ⊗ ▷⊸)`

where `⟜◁ : I → a ⊗ a` opens a pair of wires out of nothing and `▷⊸ : b ⊗ b → I` closes one, so the input of `R°` is where the output of `R` was. And it is a contravariant 2-functor `° : 𝓒ᵒᵖ → 𝓒`:

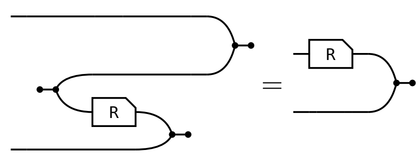
(i) `𝟙° = 𝟙` (ii) `(R S)° = S° R°` (iii) `(R ⊗ S)° = R° ⊗ S°` (iv) `R ≤ S` implies `R° ≤ S°`

## §2.1. `𝟙° = 𝟙` (snake)



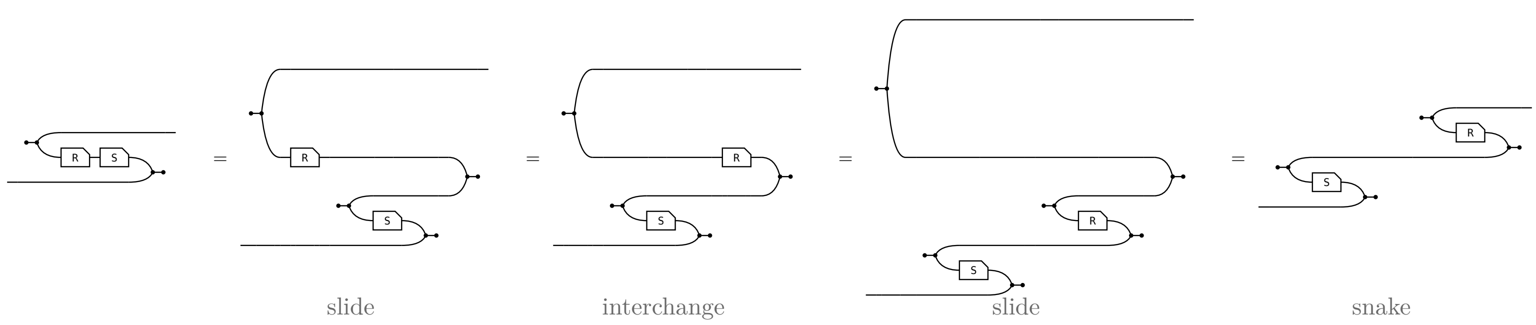

## §2.2. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:



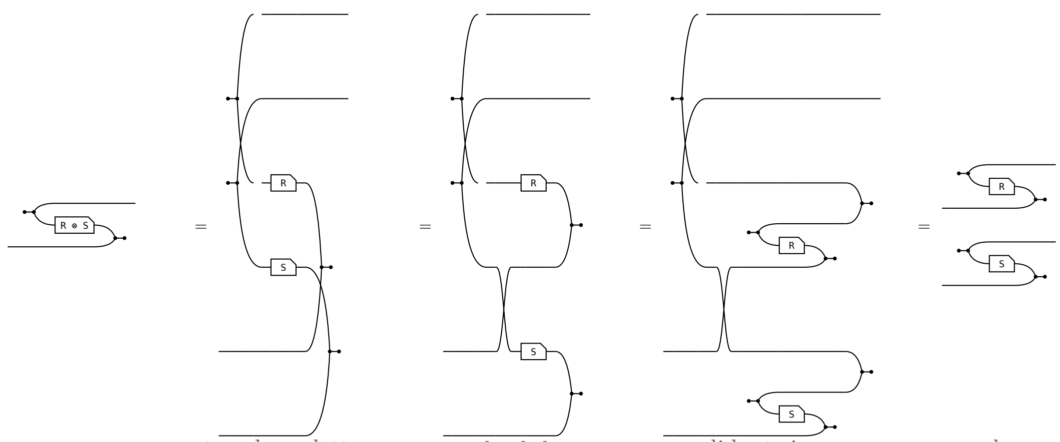

`R°` is DEFINED as the bending of `(R ⊗ 𝟙) ▷⊸`, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the `⟜◁` bends the `a` strand down and the `▷⊸` brings it back up, while the `b` strand rides through untouched. Nothing else is spent below.

## §2.3. `(R S)° = S° R°`




## §2.4. `(R ⊗ S)° = R° ⊗ S°`

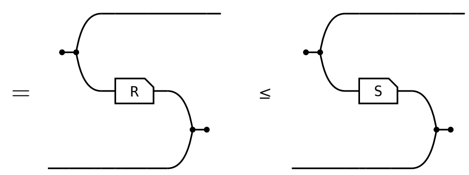
A merge at a product is two merges behind a crossing, so the pair unbends one strand at a time and the crossing is all that is left to move.




a strand per letter σ past both boxes slide, twice snake

### §2.5. $R \le S$ implies $R^\circ \le S^\circ$

$R$ sits in a frame of wires built from $\gg$ and $\otimes$, and both of those are monotone, so the box may be replaced where it stands.




$\gg$, $\otimes$ monotone

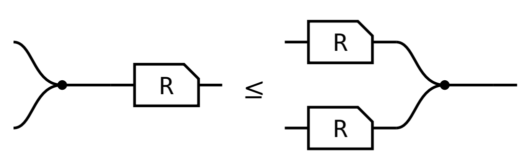
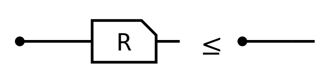
## §3. Every arrow is a lax monoid homomorphism

The merge/create mirror of the two comonoid laws opening this note — the paper's (41) and (42), p. 22, which it gets by bending them with (ii) and (iv) above. No bending is needed: expand by the unit $\mathbb{1} \le \triangleleft\ \triangleright$, duplicate the box with the comonoid law, contract by the counit $\triangleright\ \triangleleft \le \mathbb{1}$. Both halves of the adjunction, and nothing else.

In `Rel(Set)` with $R \subseteq A \times B$, where $\triangleright$ is the equality test on a pair and $\multimapinv$ produces any element at all:

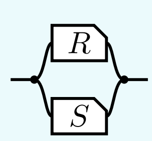
| | |
|---|---|
| $\triangleright\ R$ | $= \{((a,\ a\ ),\ b)\ :\ a\ R\ b\}$ — two equal inputs, then one run of $R$ |
| $(R \otimes R)\ \triangleright$ | $= \{((a,\ a'),\ b)\ :\ a\ R\ b$ and $a'\ R\ b\}$ — a run on each input, then the results merged |
| $\multimapinv\ R$ | $= \{(*,\ b)\ :\ \exists a.\ a\ R\ b\}$ — the image of $R$ |
| $\multimapinv$ | $= \{*\} \times B$ — all of $B$ |

Take $a' = a$ for the first containment; every $b$ lies in $B$ for the second. Each is strict exactly when its reverse fails — $R$ not injective, $R$ not surjective — which is what the later table on maps reads off them.

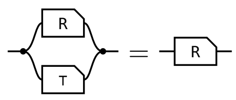
| **inequation, and what proves it** | **picture** |
|---|---|
| (41) $\triangleright\ R \le (R \otimes R)\ \triangleright$<br>$\triangleright R = \triangleright R\mathbb{1} \le \triangleright R \triangleleft\triangleright \le \triangleright\triangleleft(R{\otimes}R)\triangleright \le (R{\otimes}R)\triangleright$ — unit, lax $\triangleleft$, counit. |  |
| (42) $\multimapinv\ R \le \multimapinv$<br>$\multimapinv R = \multimapinv R\mathbb{1} \le \multimapinv R \multimap\multimapinv \le \multimapinv\multimap\multimapinv \le \multimapinv$ — the same three steps at $\mathbb{I}$. |  |

## §4. $\cap$ is a commutative idempotent monoid on every hom-set

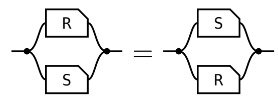
**Definition 4.1.**
The **meet** of $R,\ S\ :\ a \to b$, the paper's **convolution**, is $R \cap S := \triangleleft\ (R \otimes S)\ \triangleright$ — copy the input, run $R$ and $S$ on the two copies, merge the results — so what comes out is what both of them do.



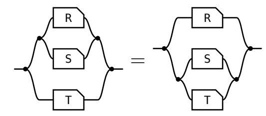
> transcribed: a definition has no statement to export
>
> On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow `⊤ = ⊸ ⟜` (the paper's Lemma 4.11).



**unit:** one half of ⊤ per end — the merge's unit law absorbs the ⟜, the copy's counit law the ⊸

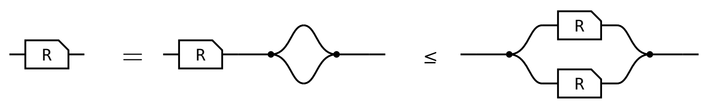


**commutative:** σ crosses R ⊗ S by naturality and is absorbed by cocommutativity and commutativity



**associative:** coassociativity and associativity; ⊗ re-brackets for nothing, being strict here

### §4.1. `R ∩ R = R`, the one that is not bookkeeping



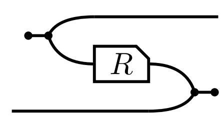
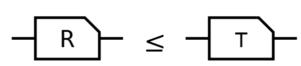
◁ ▷ = 𝟙

lax copy

The last picture is `R ∩ R` itself, so this is `R ≤ R ∩ R` and the lax copy law is the whole of it. The other direction is not proved here: `R ∩ S ≤ R` holds for every `S`, by weakening `S` to `⊤` and then collapsing `R ∩ ⊤` — which is the paper's own remaining three steps, packaged once.

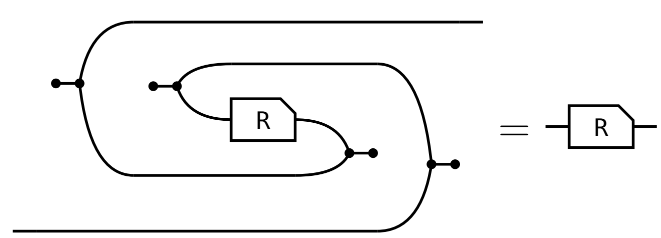
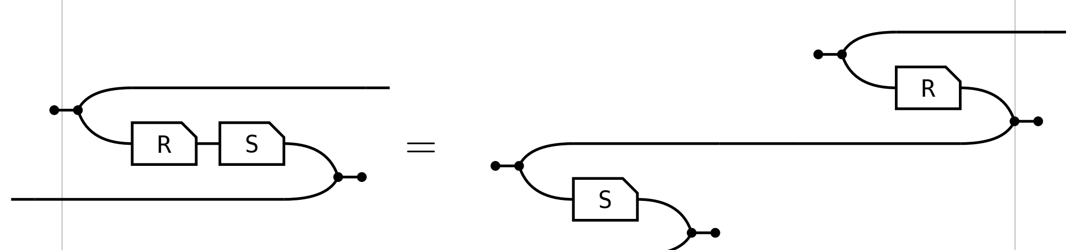
So ≤ is the order this monoid induces. `R ∩ S ≤ R` comes from the unit, and idempotency turns anything under both `S` and `T` into something under `S ∩ T`, since `R = R ∩ R ≤ S ∩ T`.

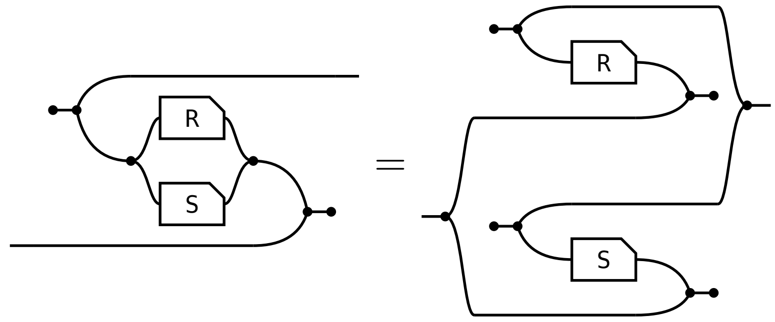
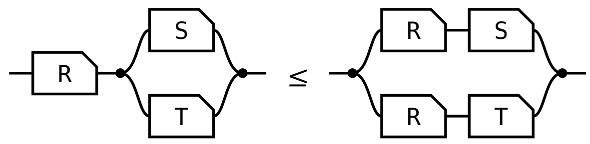
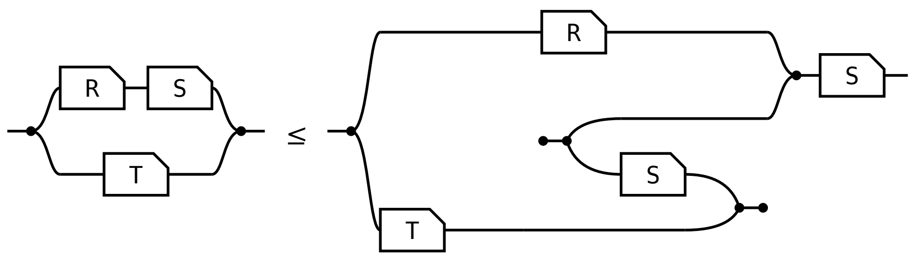
## §5. The allegory primitives are definitions here

| **primitive, and what defines it here** | **picture** |
|---|---|
| **reciprocation** `R° := bend ((R ⊗ 𝟙) ▷⊸)`, not a generator. |  |
| **containment** `R ⊑ S`, which an allegory **defines** as `R ∩ S = R`. Here ≤ is primitive and ∩ is derived, so the two directions are swapped. | the 2-cell itself |
| **maximal morphism** `⊤ := ⊸ ⟜`, free here; an allegory needs a unit object first. |  |

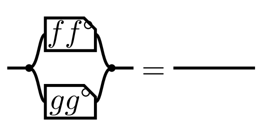
## §6. The eight axioms, and what proves each

Three of the eight — ∩ idempotent, commutative, associative — are the monoid laws above. The other five:

| **axiom, and what supplies it** | **picture** |
|---|---|
| `(R°)° = R` — the **snake**. |  |
| `(R S)° = S° R°` — mirroring the picture. |  |

| axiom, and what supplies it | picture |
|---|---|
| `(R ∩ S)° = R° ∩ S°` — the same mirroring; ◁ and ▷ are each other's mirror image. |   |
| `R (S ∩ T) ⊑ R S ∩ R T` — the lax copy law. |   |
| `R S ∩ T ⊑ (R ∩ T S°) S`, the modular law — **adjoined** as an axiom by an allegory, which has no structure to derive it from. Here Frobenius and the lax copy law give it. |   |

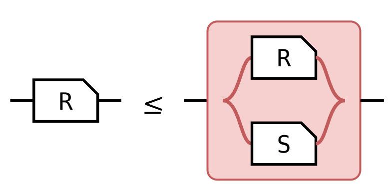
## §7. Unitary and pre-tabular — what the other direction needs

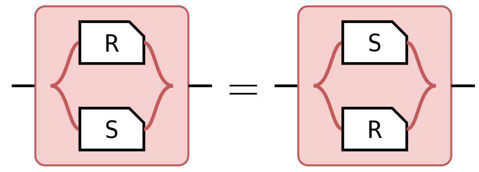
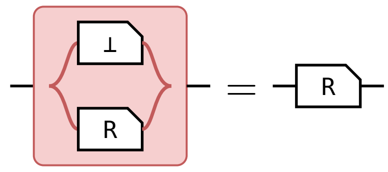
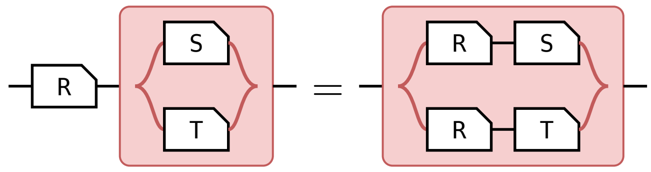
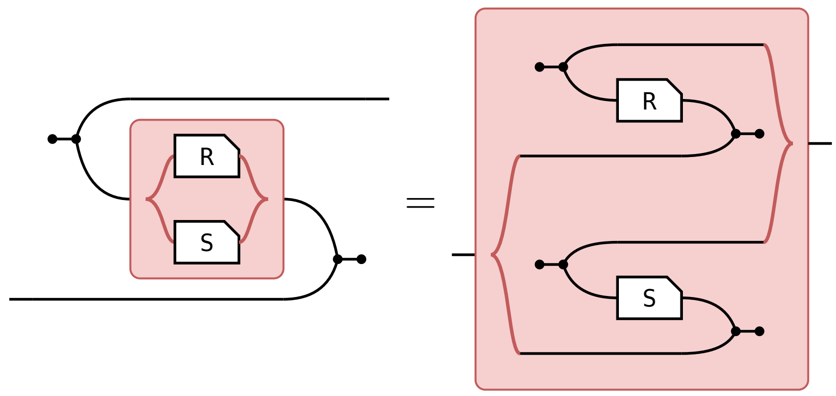
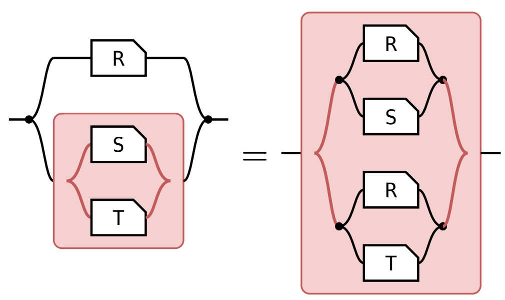
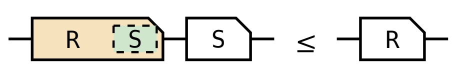
| condition | allegory | Frobenius side |
|---|---|---|
| **unitary** | `T` is a **unit** when `𝟙 T` is the largest endomorphism on it and every object carries an entire arrow to it. A unit is what makes `T` exist. | **free.** `𝕀` is the unit object, `𝟙 ≤ ⊸ ⟜` says `⊸` is entire, and `T = ⊸ ⟜` is maximal. |
| **pre-tabular** | `f, g` **tabulate** `R` when both are maps, `R = f° g`, and   `R` is **tabular** when some pair tabulates it, the allegory **pre-tabular** when every arrow lies under a tabular one. | **this is what `⊗` is.** Given unitarity it reduces to tabulating each `T`, and a tabulation of `T` is an object `n ⊗ m` with its two projections. One side derives the product from tabulations, the other posits it. |

Together: **cartesian bicategory of relations ≃ unitary pre-tabular allegory**.

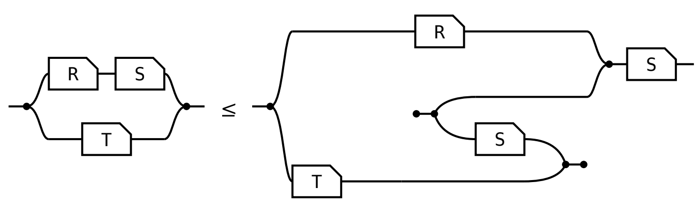
## §8. Union and residual

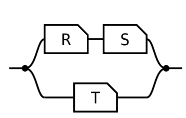
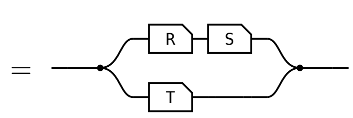
Both are past what the definition can build: a union needs a **biproduct** on top of the Frobenius structure, a residual a **second composition** with linear adjoints. A union draws as a **tape** — the rounded wrapper is the second product, and its fork and join open and close a branch a particle takes exactly one of. A residual draws as **long division**, not as its own term, which is a composition against a complement.

| statement | picture |
|---|---|
| `R ⊑ R ∪ S` `union R S := ⟨R, S⟩ [𝟙, 𝟙]`: offer both branches, then forget which was taken. |   |

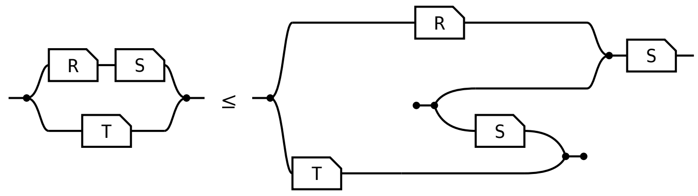
| statement | picture |
|---|---|
| $R \cup S = S \cup R$ |  |
| $\bot \cup R = R$ ⊥ routes through the zero object; there is no shape for it. |  |
| $R\ (S \cup T) = R\ S \cup R\ T$ — composition distributes over union, which $\cap$ does not. |  |
| $(R \cup S)^\circ = R^\circ \cup S^\circ$ |  |
| $R \cap (S \cup T) = (R \cap S) \cup (R \cap T)$ — what makes the whole **distributive**, and the one picture with both layers in it. |  |
| $(R\ /\ S)\ S \sqsubseteq R$ — the residual is a lower bound of the arrows it is the greatest of. |  |

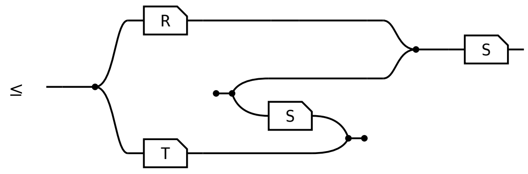
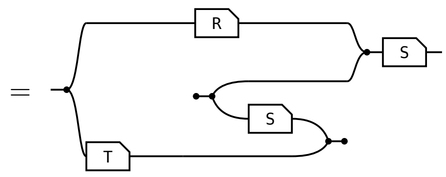
## §9. The modular law, from Frobenius

**the modular law,** $R\ S \cap T \le (R \cap T\ S^\circ)\ S$ 


| term | picture | the rule that reaches it |
|---|---|---|
| $(R\ S) \cap T$ |  | **the start:** $R\ S \cap T$. |
| $= (\lhd\ (R \otimes T))\ ((S \otimes \mathbb{1})\ \rhd)$ |  | **Factor out the shared prefix:** $(R\ S) \otimes T$ is $(R \otimes T)\ (S \otimes \mathbb{1})$. |

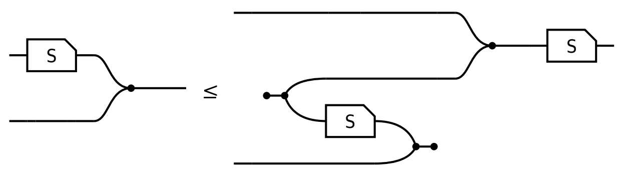
**the modular law,** `R S ∩ T ≤ (R ∩ T S°) S`

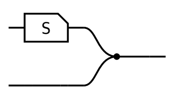
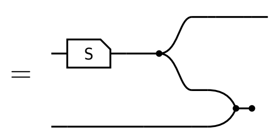
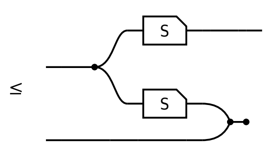
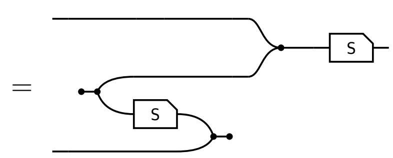



| term | picture | the rule that reaches it |
|---|---|---|
| `≤ (◁ (R ⊗ T)) ((𝟙 ⊗ S°) (▷ S))` |  | **The merge slide.** `S` slides through the merge and reappears as `S°` on the other strand. **The only inequality in the proof.** |
| `= (R ∩ (T S°)) S` |  | **Fold the prefix back in;** the copy-merge pair reads as a meet again. |

`S` occurs once on the left of the slide and twice on the right, and the lax copy law is the only law in the definition that may duplicate a box — so that is where the inequality has to be.

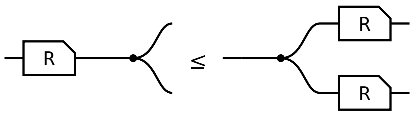
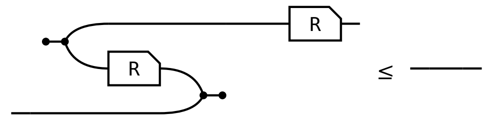
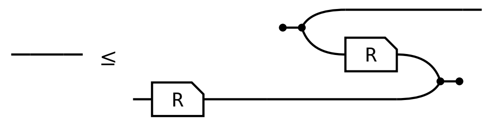
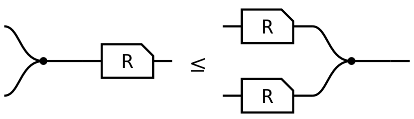
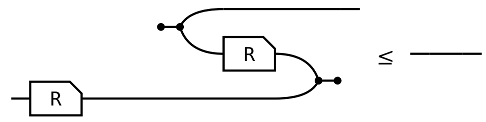
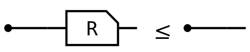
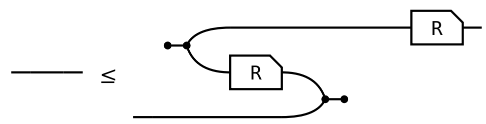
**the merge slide,** `(S ⊗ 𝟙) ▷ ≤ (𝟙 ⊗ S°) ▷ S`




| term | picture | the rule that reaches it |
|---|---|---|
| `(S ⊗ 𝟙) ▷` |  | **the start:** `(S ⊗ 𝟙) ▷`. |
| `= ((S ◁) ⊗ 𝟙) (𝟙 ⊗ ▷⊸)` |  | **Rebuild the merge** from a copy and a right bracket, the shape the lax copy law applies to. |
| `≤ ((◁ (S ⊗ S)) ⊗ 𝟙) (𝟙 ⊗ ▷⊸)` |  | **The lax copy law:** `S ◁` becomes `◁ (S ⊗ S)`, the box copied. **The whole of the modular law, in one step.** |
| `= (𝟙 ⊗ S°) (▷ S)` |  | **Reshape back,** and the surviving duplicate slides round the right bracket to become `S°`. |

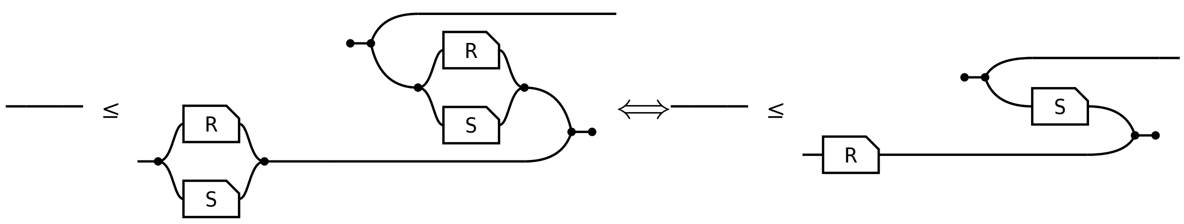
## §10. Maps

Every arrow is lax for ◁ and for ⊸ by axiom, and for ▷ and ⟜ by the section above. All four hold for **every** arrow, their REVERSES do not, and each reverse holding is a property of the arrow. The right-hand column is the **adjoint** form, which is the definition used here because it is literally the allegory's `Simple` and `Entire`; that the two forms agree is a separate theorem, not proved here.

| holds for every arrow | property | holds iff |
|---|---|---|
| R ◁ ≤ ◁ (R ⊗ R)  | (SV) **single valued** R° R ⊑ 𝟙 |  |
| R ⊸ ≤ ⊸  | (TOT) **total** 𝟙 ⊑ R R°. With **single valued**, a **map**. |  |
| ▷ R ≤ (R ⊗ R) ▷  | (INJ) **injective** R R° ⊑ 𝟙 |  |
| ⟜ R ≤ ⟜  | (SUR) **surjective** 𝟙 ⊑ R° R, that is, R° entire. |  |

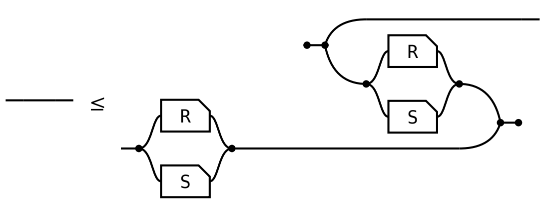
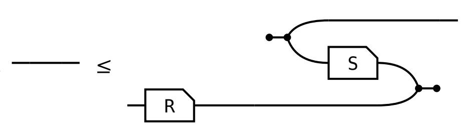
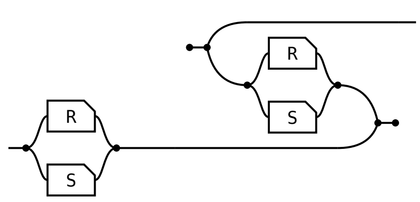
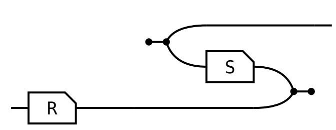
## §11. Entireness of an intersection

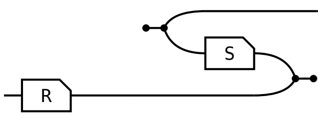
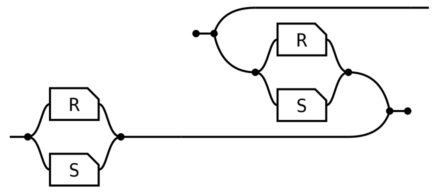
R is **entire** when 𝟙 ⊑ R R°, which is (TOT) above. When is an **intersection** entire?

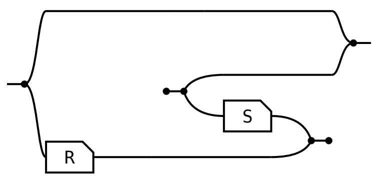
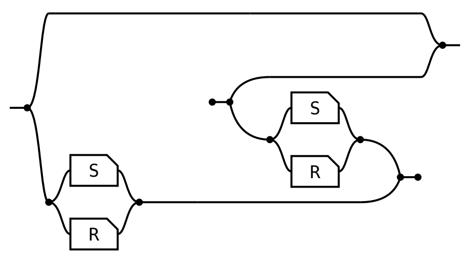
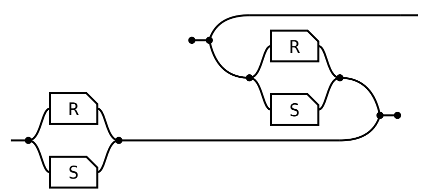


Total (R ∩ S) ↔ 𝟙 ≤ R S°

On the left R ∩ S is named **twice**; on the right R and S once each and the meet is gone. The meet is needed to **ask** whether something is entire, not to **answer** it. Where ⊑ is **defined** as X ∩ Y = X the two sides are one proposition; here ≤ is primitive, so both directions are real steps — and they cost very different things.

| ⟹ given R ∩ S entire,  show  | | |
|---|---|---|
| **term** | **picture** | **the rule that reaches it** |
| 𝟙 | —— | **the start:** 𝟙. |
| ≤ (R ∩ S) (R ∩ S)° | ≤  | **the hypothesis:** 𝟙 ≤ P P° at P = R ∩ S. |
| ≤ R S° | ≤  | **monotonicity, twice.** No modular law. |

$\Longleftarrow$ given —— $\leq$  show $R \cap S$ entire, —— $\leq$ 

| term | picture | the rule that reaches it |
|---|---|---|
| $\mathbb{1}$ | —— | **the start:** $\mathbb{1}$. |
| $\leq \mathbb{1} \cap (R\ S^\circ)$ | $\leq$  | **the hypothesis,** met with $\mathbb{1}$. |
| $= \mathbb{1} \cap ((S \cap R)\ (S \cap R)^\circ)$ | $=$  | $\mathbb{1} \cap R\ S^\circ = \mathbb{1} \cap (S \cap R)(S \cap R)^\circ$. $R\ S^\circ$ relates $a$ to $a'$ when some $b$ has $a\ R\ b$ and $a'\ S\ b$; the $\mathbb{1}\ \cap$ sets $a' = a$, and it reads “some $b$ has $a\ R\ b$ and $a\ S\ b$” — one arrow of $R \cap S$. Two arrows become one, and that is what needs the modular law. |
| $\leq (R \cap S)\ (R \cap S)^\circ$ | $\leq$  | **$X \cap Y \leq Y$,** then $S \cap R = R \cap S$. |

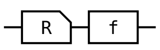
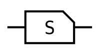
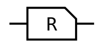
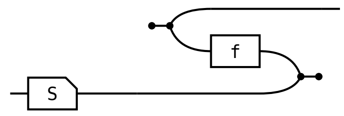
## §12. The gaps

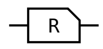
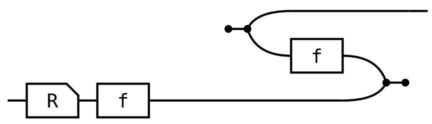
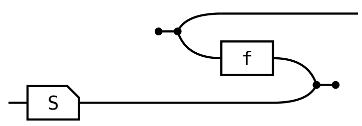
- All eight axioms are theorems of the definition, the modular identity included.
- $\rhd\lhd\leq\mathbb{1}$ and ⟜⊸$\leq\mathbb{1}$ are never used. They are there to make $\lhd \dashv \rhd$ and ⊸ $\dashv$ ⟜ genuine adjunctions, which is what pins $\rhd = \lhd^\circ$ and ⟜ $=$ ⊸$^\circ$.
- Unproved: the merge and unit forms of the lax laws, and the agreement of the comonoid and adjoint forms of the map conditions. Both trace to the one omission above.
- The bridge runs one way only: a cartesian bicategory carries $\otimes$, an allegory carries nothing of the kind, which is what the two conditions supply.
- Everything here is a theorem OF the axioms. That relations satisfy them is not checked here.

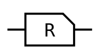
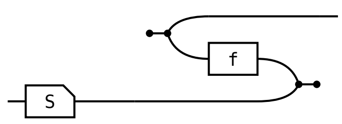
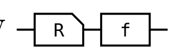
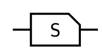
| construct | counterpart | status |
|---|---|---|
| **tabulation** | none | $\top =$ ⊸ ⟜ is all the tabular content there is. |
| $\cup$, $\bot$ | a **biproduct** $\oplus$ on top of the Frobenius structure | built, and drawn above. The calculus itself has no union. |
| $R\ /\ S$, complement | a **second composition** with linear adjoints | built, and drawn above, but axiomatised only: nothing instantiates the complement. |
| $\Lambda$, power transpose | none | open. No power object anywhere in this tower. |
| allegory $\Longrightarrow$ cartesian bicategory | rebuild $\otimes$ from tabulations | deliberately not built. |

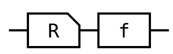
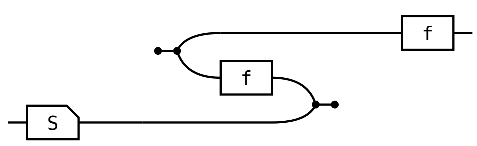
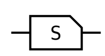
## §13. The shunting rule

A map moves from one side of a containment to the other at the cost of its converse, `R f ≤ S ⟺ R ≤ S f°`. It is how a function gets out of the way so the relation underneath can be reasoned about. Each direction spends exactly one half of **map** — never both.

**shunting right, ⟹** given  ≤  show  ≤ 

| **term** | **picture** | **the rule that reaches it** |
|---|---|---|
| `R` |  | **the start:** `R`. |
| `≤ R (f f°)` | ≤  | **total:** `𝟙 ≤ f f°`, so `f f°` may be inserted after `R`. |
| `≤ S f°` | ≤  | Re-bracket to `(R f) f°` and apply the hypothesis. |

**shunting right, ⟸** given  ≤  show  ≤ 

| **term** | **picture** | **the rule that reaches it** |
|---|---|---|
| `R f` |  | **the start:** `R f`. |
| `≤ S (f° f)` | ≤  | Apply the hypothesis on the left; the two `f`s are now adjacent. |
| `≤ S` | ≤  | **single valued:** `f° f ≤ 𝟙`, so the pair cancels. |

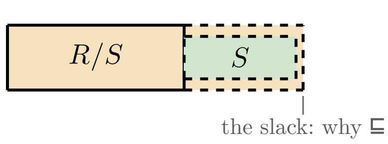
Shunting left is the mirror, same shape, map on the other side throughout. What the rule is, is an adjunction: `f ⊣ f°`, with **total** the unit and **single valued** the counit, so "`f` is a map" and "`f` has a right adjoint, namely `f°`" are one statement.

## §14. Division

Division is drawn and negation is not, because negation is not how programs get specified: the chapters where they are — greedy, thinning, dynamic programming, Horner, knapsack, paragraph formatting, string edit, bracketing, compression — use local completeness, division and power objects, and no Boolean structure at all. That also settles which picture of `/` is right. As a term a residual is a composition against a complement, which makes every homset Boolean; division asks for nothing of the kind, only the Galois connection `T ⊑ R/S ⟺ T S ⊑ R`, which `Rel(Set)` satisfies with a bare ∀, `(R / S) x y ≜ ∀z. S y z → R x z`.

**Long division.** `R / S` is how much of `R` you can lay down before `S` still fits in what is left. Read the bar as a composite: `R / S` first, then `S`, the two together inside `R`. Solid is what the equation pins — `R`'s left edge, `S`'s left edge, and the stretch before `S`. Dashed is the slack: nothing to the right of `S` is determined by `T S ⊑ R`, which is why the cancel law is `(R/S) S ⊑ R` and not an equality, and why the slack is a hairline — `R / S` is the **largest** such quotient.

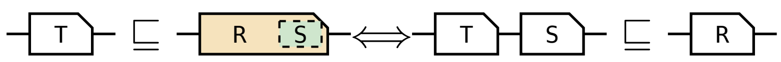
It is a **measurement**, not a containment: `R : a → c` and `S : b → c` are not in the same hom-set, so what sits inside `R` is the composite `(R/S) S`. Pushed too far the metaphor breaks in the usual place, `⊥ / ⊥ = ⊤`.




transcribed — the universal property has a statement, this metaphor does not

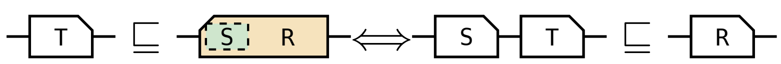
**the ten laws, their pictures, and — where the metaphor reaches — the same law as long division**

**Division is the right adjoint of composition:** $T \sqsubseteq R/S \iff T\ S \sqsubseteq R$



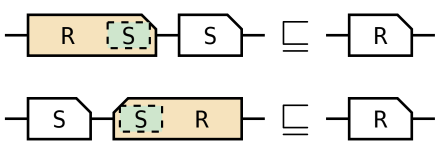

$T \sqsubseteq S\backslash R \iff S\ T \sqsubseteq R$

$S\backslash R \triangleq (R^\circ/S^\circ)^\circ$: left division is right division conjugated by the converse, not a second primitive.




**cancel:** $(R/S)\ S \sqsubseteq R$, and $S\ (S\backslash R) \sqsubseteq R$

The counit of each adjunction — the law the figure above draws.

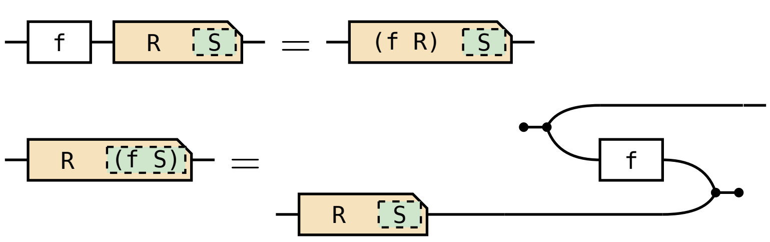



**associate:** $R/(S_1\ S_2) = (R/S_2)/S_1$, and $(S\ T)\backslash R = T\backslash(S\backslash R)$

Dividing by a composite one factor at a time; the bar shows what the label on the right hides — two tiles laid where there was one.

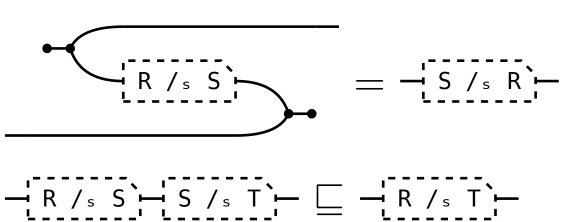

**maps:** $f\ (R/S) = (f\ R)/S$, and $R/(f\ S) = (R/S)\ f^\circ$

The same three pieces bracketed two ways, which is the licence to write $f\ R\ /\ S$; then a map moves out of a denominator and reappears as $f^\circ$ outside the box. Both are the shunting rule, spent twice.




**symmetric division:** $(R/_sS)^\circ = S/_sR$, and $(R/_sS)(S/_sT) \sqsubseteq R/_sT$

$R/_sS \triangleq (R/S) \cap (S/R)^\circ$. The meet is inside the definition, not the statement, so it stays under the label: $/_s$ is converse-symmetric and transitive, like an equality of columns.




Ten laws, ten pictures, and not one shows a generator: $\cap$, $\cup$, $^\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.